

FREMONT, NE, USA

WMO: 725564

Lat: 41.449N

Lon: 96.520W

Elev: 1203

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 04924

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.9	(c) 1.2	(d) -9.7	(e) 3.4	(f) -0.9	(g) -6.0	(h) 4.2	(i) 3.2	(j) 32.3	(k) 23.8	(l) 28.5	(m) 25.3	(n) 6.9	(o) 290	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.9	(c) 93.5	(d) 74.4	(e) 90.8	(f) 73.6	(g) 88.3	(h) 72.5	(i) 78.3	(j) 88.2	(k) 76.6	(l) 86.4	(m) 74.9	(n) 84.6	(o) 10.8	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	138.3	84.9	73.2	129.1	82.8	72.1	124.2	81.7	42.7	87.9	40.8	86.5	39.3	84.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.5	(b) 23.1	(c) 19.6	(e) -11.7	(f) 97.9	(g) 3.7	(h) 2.7	(i) -14.3	(j) 99.8	(k) -16.5	(l) 101.4	(m) -18.6	(n) 102.9	(o) -21.3	(p) 104.9		
(4) 26.5	(5) 23.1	(6) 19.6	(7) -12.3	(8) 80.8	(9) 3.5	(10) 1.6	(11) -14.8	(12) 81.9	(13) -16.8	(14) 82.8	(15) -18.8	(16) 83.7	(17) -21.3	(18) 84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	24.2	27.6	39.5	51.2	62.1	72.2	76.6	73.5	66.6	53.4	39.5	27.3	(6)
(7)		DBStd	20.81	11.98	11.81	12.15	9.96	8.89	6.87	6.10	5.69	8.44	9.59	10.27	11.39	(7)
(8)		HDD50	3028	801	629	367	104	10	0	0	2	74	338	703		(8)
(9)		HDD65	6113	1266	1047	793	425	163	18	3	6	82	378	764	1168	(9)
(10)		CDD50	3493	0	2	41	139	386	666	826	728	501	179	24	1	(10)
(11)		CDD65	1100	0	0	2	10	74	234	363	269	131	17	0	0	(11)
(12)		CDH74	10930	0	0	30	161	764	2327	3802	2377	1274	189	6	0	(12)
(13)		CDH80	4107	0	0	5	39	245	874	1603	849	446	46	0	0	(13)
(14)	Wind	WSAvg	8.5	8.8	8.9	9.8	10.6	9.4	8.4	6.8	6.2	7.6	8.6	8.8	8.6	(14)
(15)	Precipitation	PrecAvg	30.2	0.7	0.9	1.6	3.0	4.7	4.9	3.5	3.9	2.7	2.0	1.4	0.9	(15)
(16)		PrecMax	40.4	1.7	2.5	3.5	5.1	8.6	10.8	10.6	11.5	7.1	5.6	3.0	2.4	(16)
(17)		PrecMin	20.2	0.2	0.1	0.0	0.5	1.3	1.3	0.7	0.5	0.6	0.1	0.0	0.0	(17)
(18)		PrecStd	5.3	0.4	0.6	1.0	1.0	2.0	2.3	2.2	2.6	1.6	1.4	0.9	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	64.2	77.6	85.7	91.4	95.1	98.6	96.3	93.0	86.1	72.9	58.8	(19)
MCWB			43.6	50.0	57.2	62.1	67.8	73.2	74.2	73.0	72.3	66.8	56.5	50.3	(20)	
2%		DB	49.8	55.0	71.3	78.8	86.2	91.1	94.8	91.0	89.6	79.1	66.2	52.3	(21)	
		MCWB	41.5	45.6	55.7	59.4	66.9	72.2	75.5	75.2	70.8	62.1	53.7	44.2	(22)	
5%		DB	44.7	49.7	64.2	73.0	81.9	88.2	91.1	88.1	85.7	73.9	61.1	46.5	(23)	
		MCWB	38.2	42.4	53.0	56.8	64.1	71.1	75.7	74.1	69.9	59.6	50.5	40.4	(24)	
10%		DB	39.3	44.3	58.8	68.4	78.1	85.5	88.3	84.7	81.5	70.0	55.4	42.6	(25)	
		MCWB	35.1	38.7	49.1	54.8	62.9	69.7	74.4	72.6	67.4	57.7	46.9	37.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	53.7	62.8	65.1	72.2	77.8	81.1	80.1	76.0	69.6	59.7	52.6	(27)
MCD B			54.7	58.5	71.2	78.9	83.6	87.9	90.9	88.5	85.7	80.1	66.8	56.7	(28)	
2%		WB	41.5	46.4	59.0	62.1	69.6	75.4	79.0	77.8	73.8	65.6	55.5	44.9	(29)	
		MCD B	48.6	54.1	66.0	74.1	82.4	86.4	89.5	86.9	84.1	74.4	64.6	50.3	(30)	
5%		WB	38.7	42.3	53.9	59.2	67.5	73.6	77.3	75.8	72.1	62.4	51.6	41.0	(31)	
		MCD B	43.9	49.1	64.2	70.2	78.4	84.8	87.6	84.8	82.0	70.8	59.8	47.0	(32)	
10%		WB	35.6	38.8	49.1	56.2	65.1	71.8	75.6	74.1	70.1	59.0	47.6	37.4	(33)	
		MCD B	39.4	43.9	58.8	67.0	74.6	82.8	85.5	82.6	78.9	67.9	56.1	41.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.8	17.8	20.5	21.6	20.6	19.7	18.9	19.0	22.4	22.3	20.0	17.0	(35)
MCD BR			24.8	25.5	30.1	29.3	25.8	22.2	21.1	22.5	25.3	28.1	28.2	23.8	(36)	
MCWBR			17.2	16.7	16.9	14.8	11.5	9.5	9.1	9.8	10.6	14.0	16.6	16.8	(37)	
5% WB		MCD BR	23.4	23.8	26.7	24.5	22.0	19.4	18.7	18.7	21.3	22.8	25.3	22.3	(38)	
		MCWBR	16.7	16.1	15.8	14.0	11.1	10.1	9.8	10.0	10.6	13.5	16.5	16.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.310	0.330	0.390	0.413	0.419	0.427	0.419	0.378	0.330	0.296	0.284	(40)	
taud		2.442	2.392	2.391	2.236	2.245	2.290	2.269	2.291	2.361	2.483	2.502	2.453	(41)		
Ebn at Noon		279	285	292	279	274	272	268	267	272	275	272	267	(42)		
Edh at Noon		24	30	34	43	43	42	42	40	35	27	23	22	(43)		
(44)	All-Sky Solar Radiation	RadAvg	636	934	1209	1561	1841	2054	2070	1777	1459	1025	718	534	(44)	
(45)		RadStd	61	55	108	110	136	153	128	122	93	118	47	53	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page